NETWORK SECURITY

Page 1 : Introduction to Network Security

★ 1. What is Network Security ?

Network Security refers to the process of protecting computer networks, devices and data from unauthorized access, misuse, modification, disruption or destruction.
It ensures that data is transmitted securely over the network and is available only to authorized users.

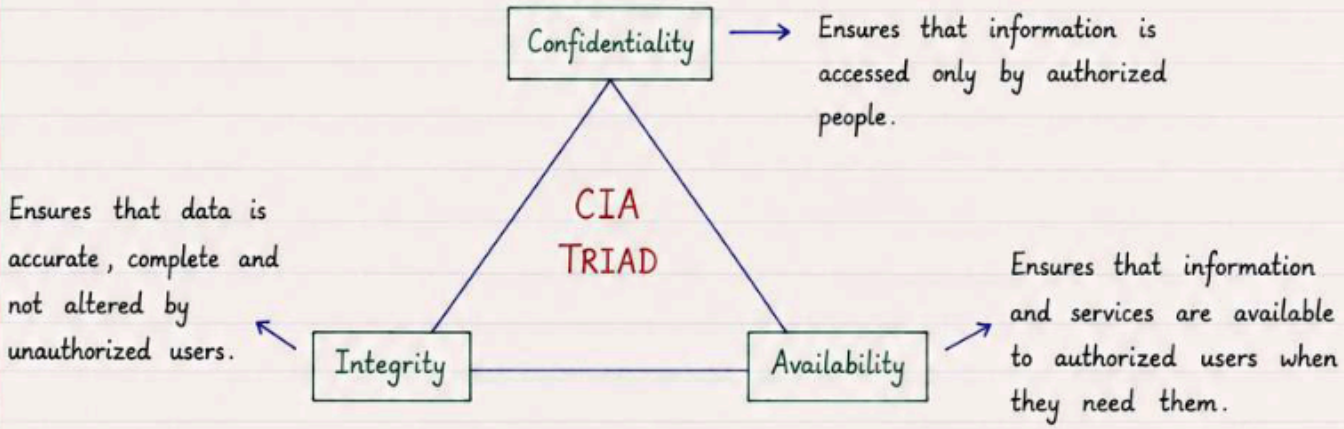
★2. Why it Matters ?

- Data travels over shared medium.
- Networks are exposed to many threats.
- Attackers can steal, modify or block sensitive information.
- Strong security protects individuals, organizations and systems.

> CIA Triad is the foundation of Network Security.

★ 3. The CIA Triad (Recap)

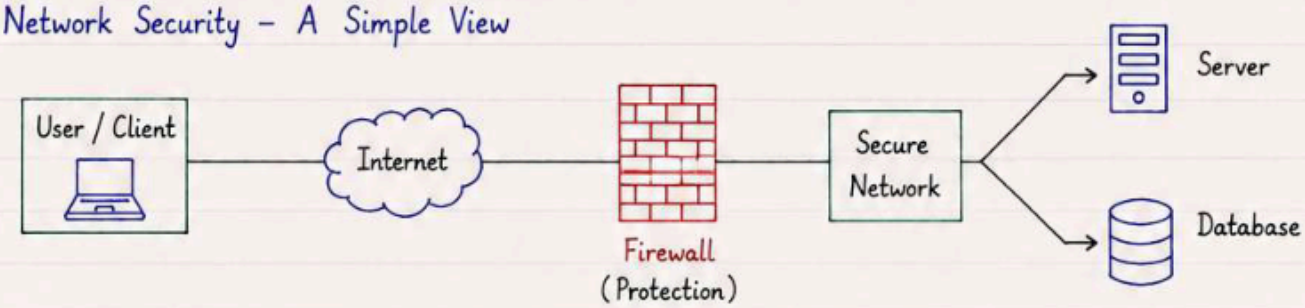
The CIA Triad represents the three core principles of information security.



★. CIA Triad in a Network Context

- Confidentiality : Protect data in transit (encryption), control access to network resources.
- Integrity : Ensure data is not modified during transmission (hashing, checksums, digital signatures).
- Availability : Keep network services and resources available (redundancy, backups, load balancing).

5. Network Security – A Simple View



> Network Security sits between users and resources to allow safe, authorized and reliable communication.

Network Security

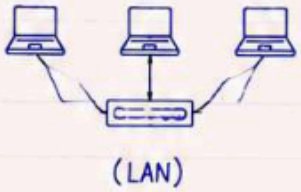
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Page 2 : Types of Networks & Common Threats

★ 1. Types of Networks (Quick Recap)

LAN (Local Area Network)

- Covers a small area like home, office, building or campus.
- High speed, low latency.
- Example : Office LAN



(LAN)

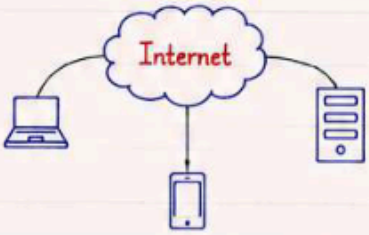
WAN (Wide Area Network)

- Covers a large geographic area like city, country or world.
- Connects multiple LANs.
- Example : Bank Network



Internet (Global Network)

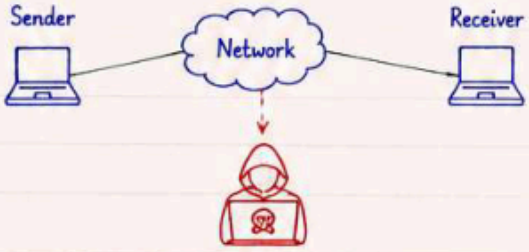
- The largest network of networks.
- Uses standard protocols (TCP/IP).
- Allows global communication.



★ 2. Common Threats in Networks

1. Eavesdropping

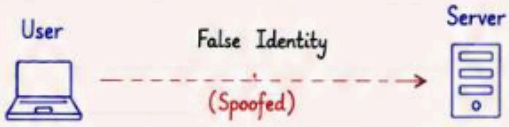
Attackers listen to network traffic to steal sensitive information.



Example :
Sniffing Wi-Fi traffic in an open network.

2. Spoofing

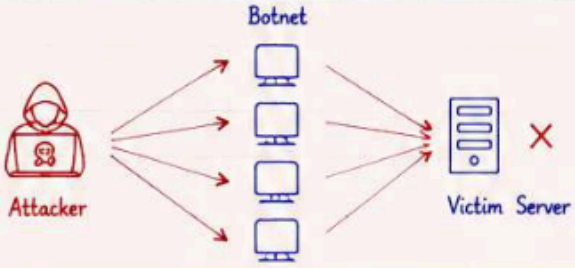
Attacker pretends to be a trusted entity to gain unauthorized access.



Example :
IP Spoofing, Email Spoofing, DNS Spoofing.

3. DoS / DDoS Attack

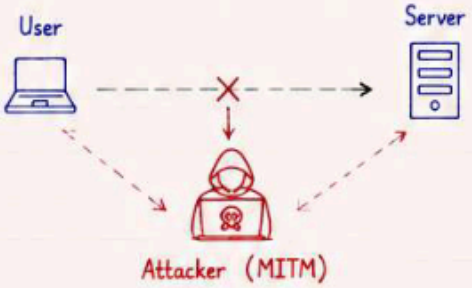
Attacker floods the network or server with traffic to make it unavailable.



Example :
DDoS attacks on websites using botnets.

4. Man-in-the-Middle (MITM)

Attacker sits between two parties to intercept and modify communication.



Example :
Fake Wi-Fi hotspot, SSL stripping.

★ Key Takeaway

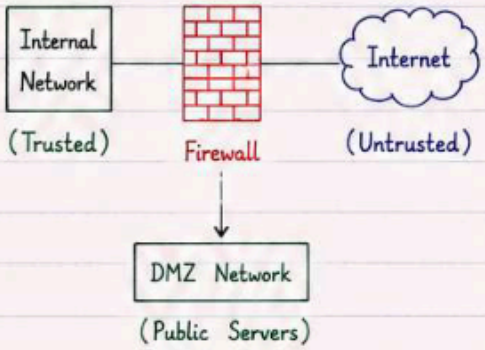
Understanding network types and common threats helps us design and implement strong security controls to protect our data and systems.

Network Security

Page 3 : Firewalls

★ 1. What is a Firewall ?

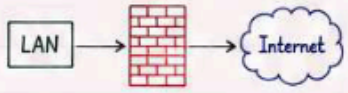
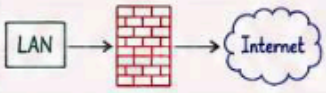
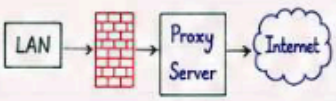
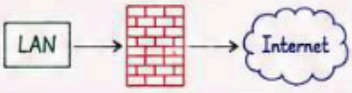
A Firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules.



Purpose :

- Allow authorized traffic
- Block unauthorized traffic
- Protect internal network from external threats

★ 2. Types of Firewalls

(1) Packet-Filtering Firewall	(2) Stateful Inspection Firewall	(3) Proxy Firewall	(4) Next-Gen Firewall (NGFW)
• Works at Network layer (Layer 3). • Filters packets based on IP addresses, ports and protocols. • Fast but less secure.	• Tracks the state of active connections. • Allows packets that are part of a valid connection. • More secure than packet-filtering.	• Works at Application layer (Layer 7). • Acts as an intermediary between client and server. • Hides internal network.	• Combines traditional firewall with advanced features. • Includes IDS/IPS, Application control, SSL inspection, etc. • Deep packet inspection.
			

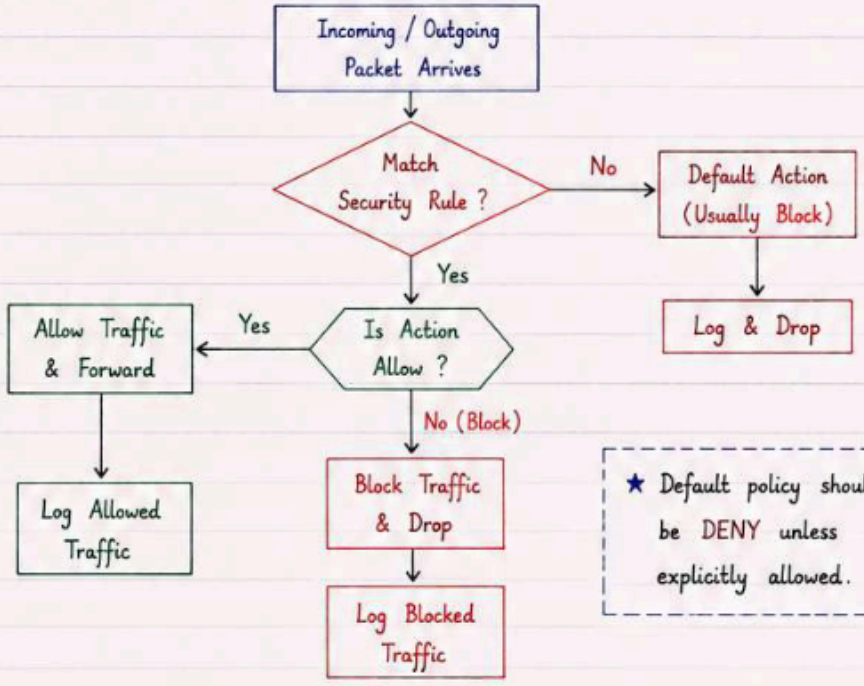
Note : Firewall is the first line of defense between trusted internal network and untrusted external network.

3. How Firewall Decides to Allow or Block Traffic ?

Firewall uses a set of rules to inspect each packet and decides whether to Allow or Block.

Information checked in a packet :

1. Source IP Address
2. Destination IP Address
3. Source Port
4. Destination Port
5. Protocol (TCP/UDP/ICMP)
6. Interface
7. Connection State
8. Application (in NGFW)



★ Example Rule :
Allow TCP traffic from 192.168.1.0/24 to any destination on port 80 (HTTP). Block all other traffic.

★ Default policy should be DENY unless explicitly allowed.

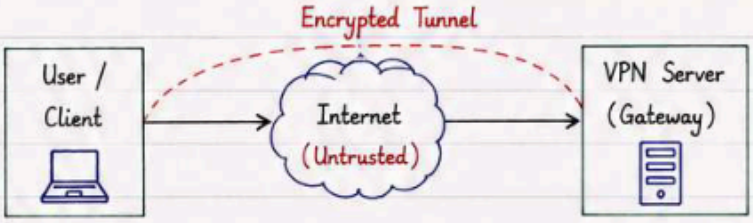
NETWORK SECURITY

Page 4 : VPN (Virtual Private Network)

★ 1. What is a VPN ?

A VPN (Virtual Private Network) creates a secure tunnel over an untrusted network (usually the Internet) to connect remote users or networks safely. It encrypts data so that only the intended parties can read it.

1.1 How VPN Works (High Level)



Data is encrypted at the client and decrypted only at the VPN server.

★ 2. Types of VPN

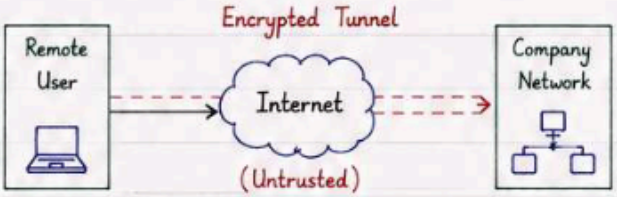
(1) Site-to-Site VPN

- Connects entire networks together.
- Used between offices, branches or data centers.
- Devices in both networks act like they are on the same private network.



(2) Remote Access VPN

- Connects individual users to a private network.
- Used by employees working from home or while traveling.
- User's device becomes part of the private network.



★ 3. Common VPN Protocols

(1) IPSec VPN

- Works at the network layer (Layer 3).
- Provides authentication, encryption and integrity.
- Suitable for site-to-site VPN.
- Uses AH (Authentication Header) and ESP (Encapsulating Security Payload).

(2) SSL / TLS VPN

- Works at the application layer (Layer 7).
- Uses SSL / TLS to create secure connections.
- Suitable for remote access VPN.
- Commonly used through web browsers or dedicated VPN clients.

★. Key Benefits of VPN

- Encrypts data and keeps it confidential.
- Protects data integrity during transmission.
- Allows secure access over public networks.
- Cost-effective compared to leased private lines.
- Supports remote work and distributed teams.

> **Note :**
> VPN does not make the Internet itself secure. It creates a secure tunnel for your data between two trusted endpoints.

5. VPN Use Cases

Business	Remote Work	Travel	Partners / Vendors
• Connect branch offices. • Connect to corporate resources securely.	• Employees access internal systems from home. • Secure file sharing and communication.	• Safe access to corporate network while traveling. • Protects from insecure Wi-Fi networks.	• Secure access for third parties to specific internal resources.

Network Security

Page 5 : IDS vs IPS

★ 1. What are IDS and IPS ?

IDS (Intrusion Detection System) :

Monitors network or system activities and detects suspicious or malicious behavior.

IPS (Intrusion Prevention System) :

Not only detects but also blocks or prevents the detected threats in real-time.

2. IDS vs IPS (Key Differences)

Feature	IDS	IPS
Full Form	Intrusion Detection System	Intrusion Prevention System
Primary Goal	Detect threats and alert	Detect and prevent threats
Action	Passive (Monitoring)	Active (Blocking)
Placement	Out-of-band (Mirror Traffic)	In-line (Traffic Path)
Response Time	After the attack	During the attack
Impact on Traffic	No direct impact	May add slight latency
Example	Snort (IDS), Wireshark	Snort (IPS mode), Suricata IPS, Cisco IPS

★ 3. Detection Techniques

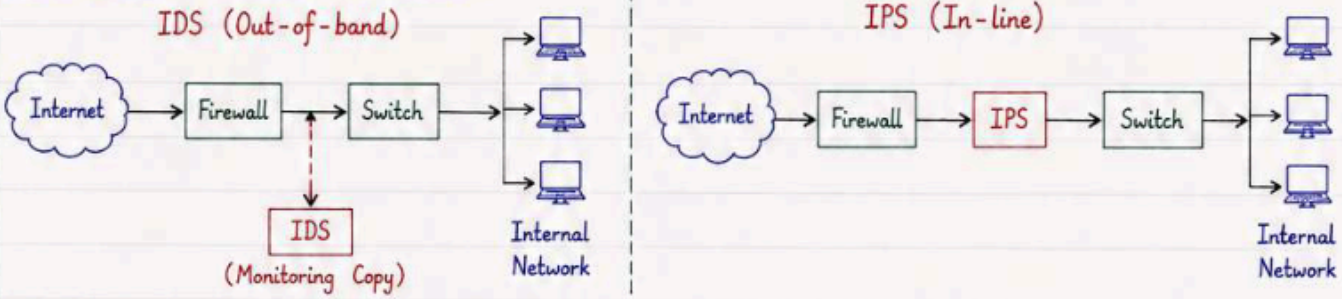
(1) Signature-Based Detection

- Uses a database of known attack patterns (signatures).
- Compares network activity with known signatures.
- Effective for known threats.
- Less effective against new / unknown attacks.

(2) Anomaly-Based Detection

- Learns normal behavior (baseline) of the network.
- Detects any deviation from normal behavior as suspicious.
- Can detect unknown / new attacks.
- May produce false positives.

4. IDS vs IPS Placement in Network



IDS only alerts the admin. IPS can take action automatically to block or drop malicious traffic.

5. Common Use Cases

Use IDS when :

- You want visibility into attacks.
- You are in monitoring or learning phase.
- You want logs and alerts for analysis.
- Minimum impact on network is required.

Use IPS when :

- You want real-time protection.
- You want to stop threats automatically.
- You are protecting critical systems.
- You need compliance and strong security enforcement.

★ Key Takeaway : IDS detects and informs.
IPS detects and prevents.

NETWORK SECURITY

Page 6 : Ports & Protocols

★ 1. Common Ports (Well-Known Ports)

Port	Protocol	Service
21	TCP	FTP (File Transfer)
22	TCP	SSH (Secure Shell)
23	TCP	Telnet (Remote Access)
25	TCP	SMTP (Email Sending)
53	UDP/TCP	DNS (Domain Name System)
67	UDP	DHCP (Server)
68	UDP	DHCP (Client)
80	TCP	HTTP (Web)
110	TCP	POP3 (Email Receiving)
143	TCP	IMAP (Email Receiving)
443	TCP	HTTPS (Secure Web)
3389	TCP	RDP (Remote Desktop)

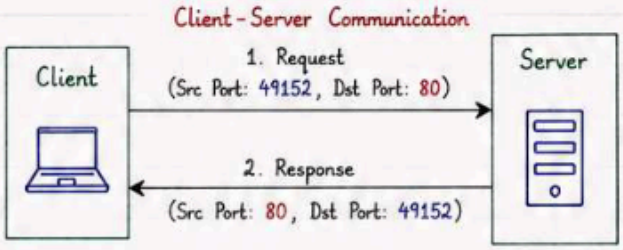
2. TCP vs UDP (Quick Comparison)

Feature	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-Oriented	Connectionless
Reliability	Reliable (Acknowledgment)	Unreliable (No ACK)
Data Delivery	In-order delivery	No order guarantee
Speed	Slower (More Overhead)	Faster (Less Overhead)
Error Checking	Yes	Yes (Checksum only)
Flow Control	Yes	No
Examples	HTTP, FTP, SMTP, SSH	DNS, DHCP, VoIP, Video Streaming, Online Gaming

3. Why Port Security Matters ?

- Open ports act like doors to your system.
- Unnecessary open ports increase attack surface.
- Attackers scan open ports to find vulnerabilities.
- Proper port management helps protect the network.

★ 4. How Port Communication Works ?



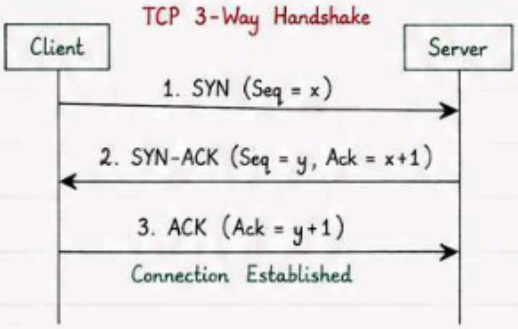
Client uses an ephemeral (random high) port.
Server listens on a well-known port.

5. Common Protocols (Overview)

- HTTP : Web browsing (port 80)
- HTTPS : Secure web browsing (port 443)
- FTP : File transfer (port 21)
- SSH : Secure remote access (port 22)
- DNS : Domain name resolution (port 53)
- DHCP : IP address assignment (port 67/68)
- SMTP : Email sending (port 25)
- POP3/IMAP : Email receiving (port 110/143)
- RDP : Remote Desktop Protocol (port 3389)

6. Port States (TCP - 3 Way Handshake)

CLOSED : No connection. Port not in use.
LISTEN : Waiting for incoming connection (at server side).
SYN-SENT : Connection request sent.
SYN-RECEIVED : Connection request received.
ESTABLISHED : Connection is established.
FIN-WAIT : Connection is closing.
TIME-WAIT : Waiting for delayed packets.



3-way handshake ensures both sides are ready before actual data transfer.

★ Key Takeaway

- Know the important ports and protocols.
- Use only required ports and close unnecessary ones.
- Prefer secure protocols (HTTPS, SSH, SFTP, etc.).

Remember for Interviews

- Why is port 80 not secure ?
- Difference between TCP and UDP.
- What happens in TCP 3-way handshake ?
- Why is port scanning dangerous ?

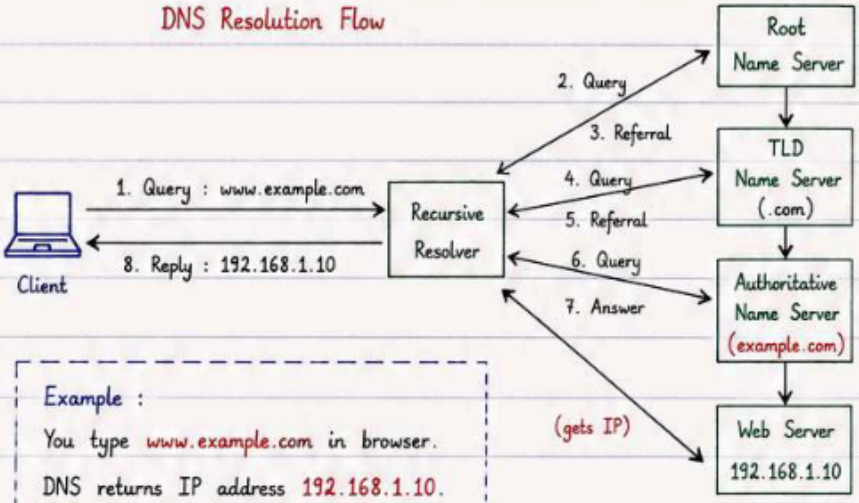
Network Security

Page 7 : DNS Security

★ 1. How DNS Works (Quick Recap)

- DNS (Domain Name System) translates human-readable domain names into IP addresses.
- It uses a hierarchy of DNS servers to resolve queries.
- Uses UDP port 53 (TCP for zone transfers and large replies).

DNS Resolution Flow



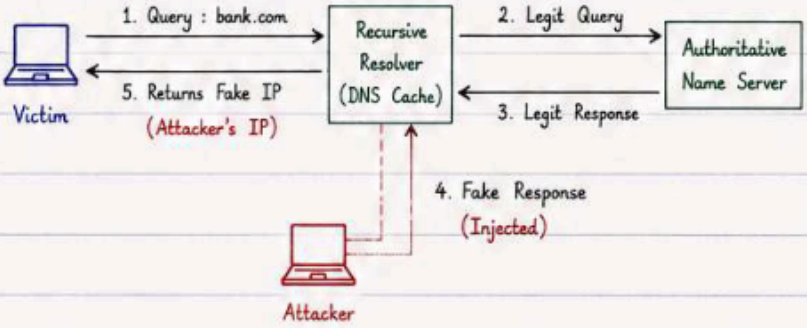
Example :
You type www.example.com in browser.
DNS returns IP address 192.168.1.10.
Browser connects to that IP.

★ 2. DNS Spoofing / Cache Poisoning

- Attacker sends fake DNS responses to DNS server or client.
- Malicious entry is stored in DNS cache.
- Victim is redirected to attacker's server instead of real server.

Impact :
- Users go to fake websites.
- Credential theft.
- Malware distribution.

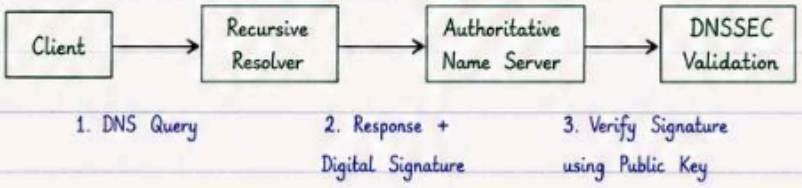
DNS Spoofing Attack (Cache Poisoning)



★ 3. DNSSEC (DNS Security Extensions)

- DNSSEC adds a layer of security to DNS.
- Uses digital signatures to verify authenticity of DNS data.
- Prevents data tampering and cache poisoning.
- Ensures data is from legitimate source.

DNSSEC Validation Flow



Benefit : Ensures integrity and authenticity of DNS responses.

★ 4. Key Takeaways

- DNS is critical for internet connectivity.
- DNS attacks can be used for phishing, malware, and data theft.
- Always use DNSSEC, secure resolvers and monitor DNS traffic.
- Implement access controls and logging on DNS servers.

Common DNS Attacks	
Attack	Description
DNS Spoofing	Faking DNS responses to redirect users.
Cache Poisoning	Injecting false data in DNS cache.
DNS Tunneling	Using DNS for covert data exfiltration.
DNS Amplification	Exploiting DNS for DDoS attacks.

NETWORK SECURITY

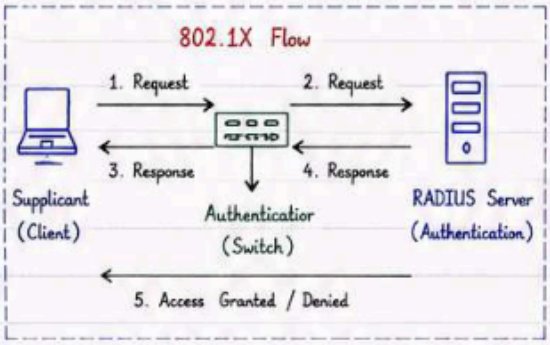
Page 8 : Network Access Control

★ 1. Authentication Methods

- Authentication verifies the identity of users or devices before granting network access.
- Prevents unauthorized access and reduces security risks.

(1) 802.1X (Port-Based Authentication)

- Standard for network access control.
- Authenticates user/device before allowing access to the network.
- Uses RADIUS server for authentication.



(2) MAC Address Filtering

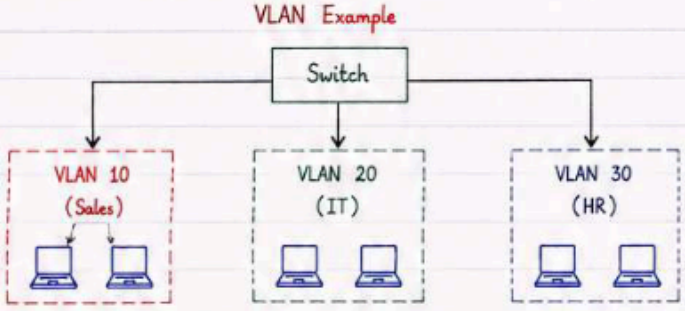
- Allows or blocks devices based on their MAC addresses.
- Simple but can be easily bypassed by spoofing.

(3) Password / Credential Based

- Users authenticate using username and password.
- Common in VPN, Wi-Fi and system logins.
- Use strong passwords and MFA for better security.

2. VLANs (Virtual Local Area Networks)

- VLANs divide a physical network into multiple logical networks.
- Helps in isolation, security and efficient traffic management.
- Devices in different VLANs cannot communicate by default.



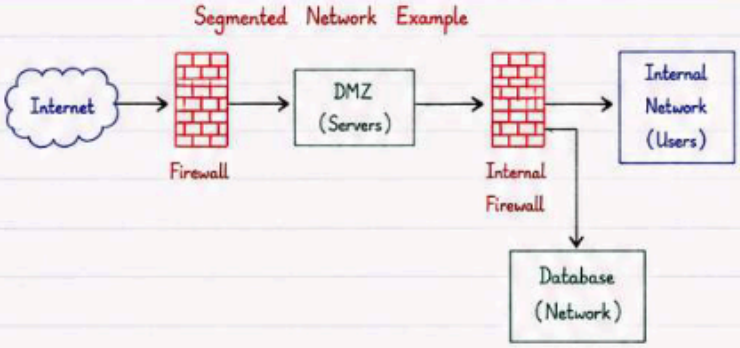
Each VLAN is a separate broadcast domain.

★ 3. Network Segmentation

- Dividing network into smaller segments or zones.
- Limits attacker movement if one segment is compromised.
- Protects critical assets and sensitive data.

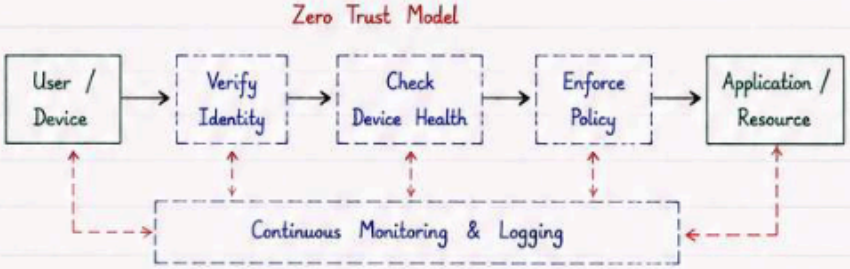
Benefits :

- Improves security.
- Reduces risk of lateral movement.
- Easier to monitor and manage.



★ 4. Zero Trust Network (ZTN)

- Never trust, always verify.
- Every user, device and application must be verified before access.
- Assume breach and minimize access privileges.
- Enforce least privilege access.
- Continuously monitor and validate.



Key Takeaways

- Authenticate every user and device.
- Use VLANs and segmentation to isolate networks.
- Apply Zero Trust principles for stronger security.

Quick Recap

Topic	Key Point
Authentication	Verify identity before granting access.
VLAN	Logical separation for security and management.
Segmentation	Divide network to reduce risk and isolate threats.
Zero Trust	Trust nothing, verify everything, least privilege.

NETWORK SECURITY

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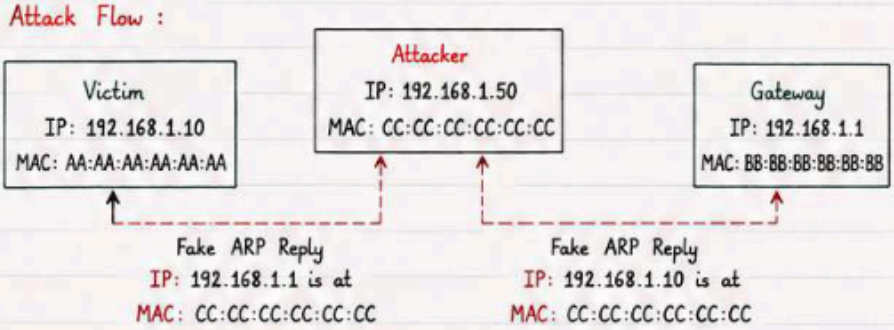
Page 9 : Common Network Attacks (Scenario-Based)

Understanding attacks helps in identifying and preventing them in real-world.

1. ARP Poisoning (ARP Spoofing)

- Attacker sends fake ARP replies to associate his MAC with IP of the gateway or another victim.
- Traffic meant for the gateway goes to attacker.
- Attacker can sniff, modify or block packets.

Attack Flow :

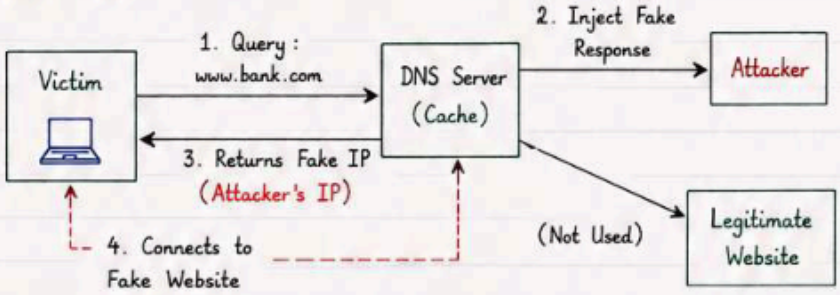


Impact :
- Packet Sniffing
- Session Hijacking
- Data Manipulation
- DoS

2. DNS Spoofing (DNS Cache Poisoning)

- Attacker injects fake DNS records into DNS server cache.
- Victim gets wrong IP address for a legitimate domain.
- Users are redirected to malicious sites.

Attack Flow :

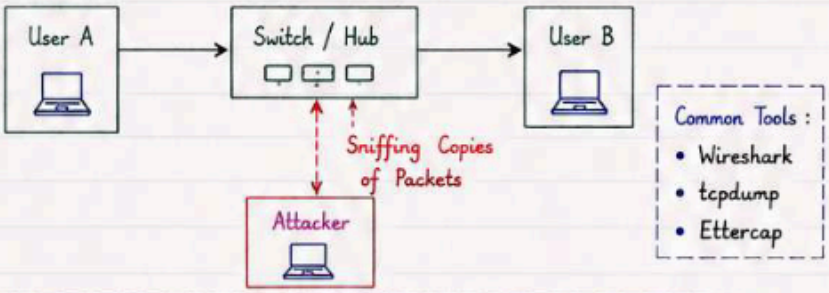


Impact :
- Phishing
- Credential Theft
- Malware Distribution

3. Packet Sniffing

- Attacker captures packets traveling over the network.
- Used to steal credentials, cookies, or sensitive data.
- More effective on unsecured networks (No encryption).

Attack Flow :

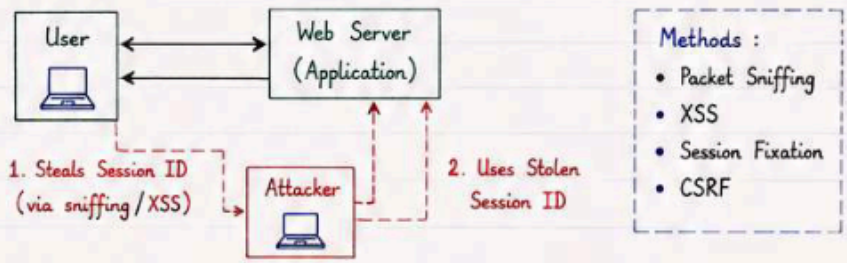


Impact :
- Data Exposure
- Credential Theft
- Session Hijacking

4. Session Hijacking

- Attacker takes over an active session between client and server.
- Done by stealing session ID / cookies or predicting tokens.
- Attacker can perform actions as the legitimate user.

Attack Flow :



Impact :
- Unauthorized Access
- Account Takeover
- Data Theft

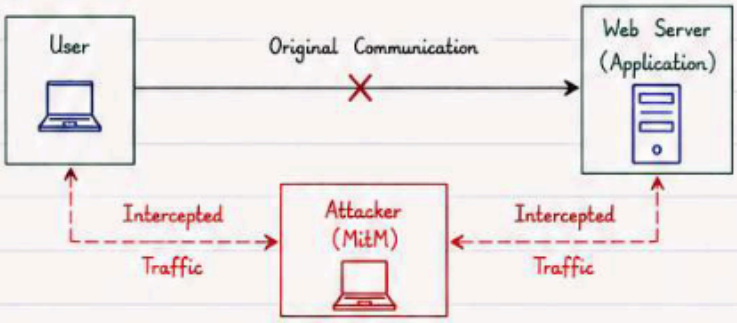
Network Security

Page 10 : Common Network Attacks (Contd.)

5. Man-in-the-Middle (MitM)

- Attacker secretly intercepts and may alter communication between two parties.
- Both parties think they are communicating directly.
- Can steal or modify sensitive data.
- Common in unsecured Wi-Fi networks.

Attack Flow :

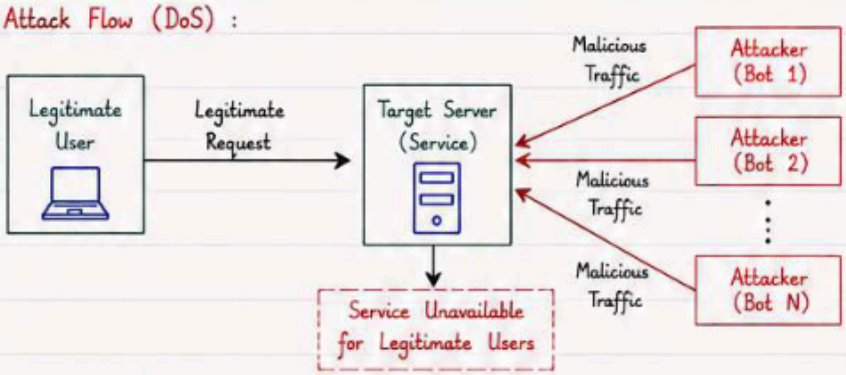


Impact :
- Data Theft
- Credential Theft
- Session Hijacking
- Data Tampering

6. Denial of Service (DoS)

- Attacker floods the target with traffic or requests.
- Exhausts resources making the service unavailable.
- Legitimate users cannot access the service.
- DDoS uses multiple systems to amplify the attack.

Attack Flow (DoS) :



Impact :
- Downtime
- Financial Loss
- Reputation Damage
- Business Disruption

7. Routing Attacks

- Attacker manipulates routing information.
- Traffic is redirected through attacker.
- Examples : IP Spoofing, Route Injection, BGP Hijacking.

Impact :
- Data Interception
- Traffic Redirection
- Service Disruption
- Loss of Integrity

8. DHCP Starvation

- Attacker exhausts DHCP IP address pool.
- Legitimate devices cannot get IP addresses.
- Network connectivity is affected.

Impact :
- Loss of Connectivity
- Service Unavailability
- Operational Disruption

9. Social Engineering Attacks

- Attacker manipulates people to reveal confidential information.
- Uses techniques like phishing email, fake calls, or impersonation.
- Targets the weakest link - humans.

Impact :
- Credential Theft
- Data Breach
- Financial Fraud
- Reputation Loss

10. Quick Comparison

Attack	Main Goal	Target	Example
ARP Spoofing	Redirect traffic	LAN	Fake ARP replies
DNS Spoofing	Redirect users	DNS	Fake DNS records
Packet Sniffing	Steal data	Network	Capture packets
Session Hijacking	Take over session	Active session	Steal session ID
DoS / DDoS	Make service down	Server / Network	Flood with traffic
MitM	Intercept & modify	Communication	Fake Wi-Fi AP
Routing Attacks	Redirect routes	Network	Route Injection
DHCP Starvation	Exhaust IP pool	DHCP Server	Fake DHCP requests
Social Engineering	Trick users	People	Phishing email

★ Key Takeaway: Understand how attacks work is the first step. The next step is to implement security controls, monitor continuously and stay one step ahead!